

Modeling with Systems and Matrices

Systems from context, matrix equations, and nonlinear systems

Directions

- Define your variables before you write any equation — say what each letter counts or measures.
- Show the system (or the matrix equation $AX = B$) you set up, then solve it.
- Check every answer in *both* original equations before you write it on the answer line.

1. A school play sells adult tickets at \$8 and student tickets at \$5. On opening night 120 tickets were sold and \$825 was collected. Let a be the number of adult tickets and s the number of student tickets sold. Write a system of equations and find how many of each kind were sold.

Answer: _____

2. A lab mixes a 20% acid solution with a 50% acid solution to make 12 L of a 30% acid solution. Let x be the number of litres of the 20% solution and y the number of litres of the 50% solution. Write a system and find how much of each solution is needed.

Answer: _____ L

3. A print shop bills two orders. Order one: 3 boxes of pens and 2 boxes of markers cost \$16. Order two: 4 boxes of pens and 5 boxes of markers cost \$33. Let x be the price of a box of pens and y the price of a box of markers, in dollars.

- (a) Write the two orders as one matrix equation $AX = B$ and compute $\det A$.
- (b) Solve the matrix equation for x and y .

Answer: _____

4. A tour company seats groups in vans and buses. 2 vans and 3 buses seat 96 passengers; 5 vans and 2 buses seat 108 passengers. Let x be the number of seats in a van and y the number of seats in a bus.

- (a) Write the coefficient matrix A and compute $\det A$.
- (b) Solve for x and y using Cramer's rule.

Answer: _____

5. A drone climbs along the curve $y = x^2 - 4x + 7$ while a support cable runs along the line $y = 2x - 1$, where x and y are both measured in metres from the base of the tower. Find every point where the drone's path meets the cable.

Answer: _____

6. A kayaker paddles 12 km downstream in 2 hours, then paddles the same 12 km back upstream in 3 hours. Let k be the kayak's speed in still water and c the speed of the current, both in km/h. Write a system and find both speeds.

Answer: _____

7. A workshop builds chairs and tables. Each chair takes 4 hours of cutting and 6 hours of finishing; each table takes 3 hours of cutting and 2 hours of finishing. This week exactly 27 cutting hours and 28 finishing hours are used. Let x be the number of chairs and y the number of tables.

- (a) Write the plan as a matrix equation and compute $\det A$.
- (b) Solve for x and y .

Answer: _____

8. A survey drone flies along the line $y = x + 1$. A circular restricted zone has boundary $x^2 + y^2 = 25$, with both coordinates in kilometres. Find every point where the flight path crosses the boundary of the zone.

Answer: _____

9. A family invests \$4000 in two accounts. One account pays 5% simple interest per year and the other pays 8% per year. After one year the two accounts together have earned \$275 in interest. Let x be the amount invested at the lower rate and y the amount invested at the higher rate, in dollars. How much went into each account?

Answer: _____

10. A caterer files two order sheets for the same event. Sheet one: 2 trays of sandwiches and 3 trays of fruit, for 12 servings. Sheet two: 4 trays of sandwiches and 6 trays of fruit, for 30 servings. Let x be the servings in a sandwich tray and y the servings in a fruit tray.

- (a) Compute $\det A$ for the coefficient matrix of this system.
- (b) Solve the system, or show that it has no solution.
- (c) Explain what your determinant tells you about whether both order sheets can be correct.

Answer: _____

11. A rectangular community garden is enclosed by 34 m of edging and covers 70 m^2 of ground. Find the two side lengths of the garden.

Answer: _____ m

12. A concert sells adult tickets at \$12, student tickets at \$8 and child tickets at \$5. In all, 200 tickets were sold for a total of \$1800, and the number of student tickets sold was exactly 2 times the number of child tickets. Let x , y and z be the numbers of adult, student and child tickets. Write a three-variable system and find how many of each were sold.

Answer: _____